

• MSC THESIS DEFENSE

Spatial Estimation of Air Quality at Unmonitored Locations Using Machine Learning

Implications for IoT Sensor Network Design in Swedish Cities

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Sparse in the north, dense in the south — the same imbalance this thesis measures

- THE PROBLEM

Between stations, nobody knows the air

- SMHI's reference network is accurate but sparse — stations can be **tens of km apart**
- The space between stations is unmonitored; air quality there is only ever a guess
- Cheap IoT sensors could fill the gap, but today they get placed by **convenience**, not analysis

This thesis builds the missing decision rule: where does a sensor actually help?

- WHY IT MATTERS

Low concentrations, real health cost

- PM2.5 and NO2 are linked to respiratory and cardiovascular disease, even at Sweden's comparatively low levels
- Studies in Stockholm, Gothenburg and Umeå show measurable attributable mortality from traffic-related exposure
- Exposure varies sharply *within* a city — a handful of national stations can't see that

Segersson et al., 2017

• WHAT'S ALREADY OUT THERE

Three ways to estimate the gap

• **IDW**

Distance-weighted average of nearby stations. No assumptions, the standard baseline.

• **LUR**

Land-Use Regression — Europe's established peer-reviewed standard, calibrated across 20 study areas (ESCAPE).

• **Machine Learning**

Captures non-linear patterns; literature shows it often beats LUR on accuracy.

But which one wins depends entirely on how you test it — and that's where this gets interesting.

• A METHODOLOGICAL TRAP

Why random testing lies to you



THE TRAP

Randomly splitting stations into train/test looks rigorous — but nearby stations look alike. The model quietly sees a preview of the answer through its neighbors.

The fix: hide a whole **neighborhood** around each test station, not just the station itself (Roberts et al., 2017). This becomes the backbone of the entire validation strategy.

- THE GAP

Nobody had chained these together

- Plenty of research estimates pollution at unmonitored spots
- Plenty of research optimizes sensor placement
- Nobody uses a **measured accuracy-vs-distance relationship**, at national scale, on public data, as the actual placement rule



THE OPENING

This thesis is the missing link between “how good is our estimate” and “so where do we put the next sensor.”

• RESEARCH QUESTIONS

Three questions, one chain

RQ1

How accurately can we estimate PM2.5 & NO2 at a station, using only the other stations?



RQ2

How does that accuracy decay with distance — and where's the cutoff?



RQ3

Using that cutoff, where should new sensors go?

● RESEARCH DESIGN

Design Science Research

PHASE	THIS THESIS
Problem & objectives	The coverage gap between SMHI stations, framed into three RQs
Design & development	Random Forest model + accuracy-distance decay curve + greedy placement algorithm
Demonstration	Applied nationally across the whole of Sweden
Evaluation	Judged by real-world coverage gained, not just model accuracy scores

The output has to work as a useful tool, not just score well as a model.

• A MODELLING DECISION

Two pollutants, two models

• **NO₂**

Traffic-sourced · sharp gradients · concentrated close to roads, fades fast with distance

• **PM_{2.5}**

Mixed regional sources · smooth gradients · spreads gradually over tens of kilometres

Different physics, so separate models throughout — merging them would blur two real patterns into one average.

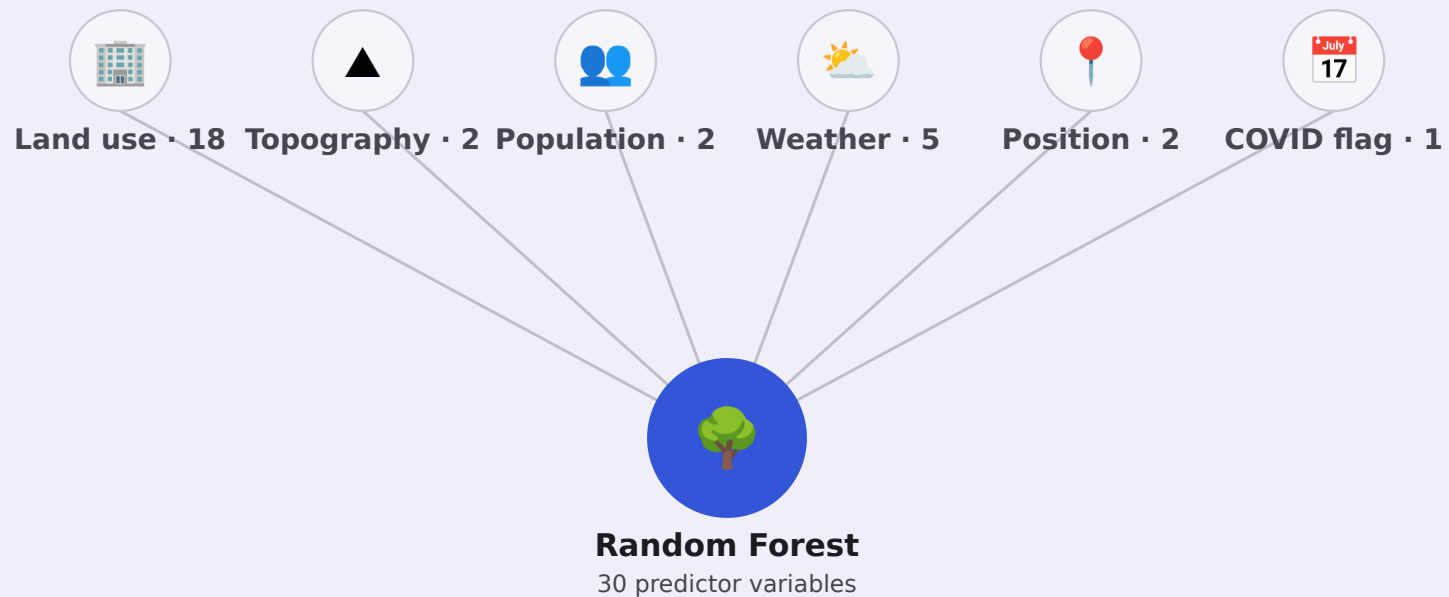
• INPUTS TO THE MODEL

30 features, five groups — each earning its place



WHY IT'S CREDIBLE

Every feature group ties back to a physical mechanism for why it should predict pollution — not a data-dump.



Land use · 18

Topography · 2

Population · 2

Weather · 5

Spatial position · 2

COVID flag · 1

- CHOOSING THE ESTIMATOR

Random Forest, deliberately

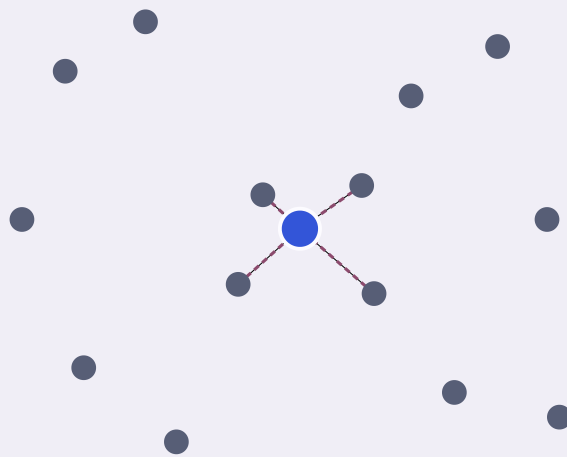
- Handles mixed feature types without assuming straight-line relationships
- Doesn't fall apart on a small, oddly-shaped dataset — only 11 usable stations
- Deep learning and hybrid Kriging models were **excluded**: they need far more training locations than Sweden's network offers

Matching model complexity to data availability — not chasing the fanciest option available.

• TESTING METHOD

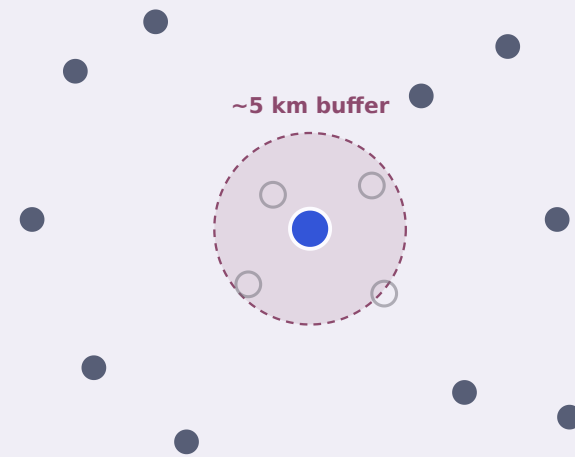
Buffered spatial leave-one-out

⚠ Random split



Near neighbors stay in training
— the model previews the answer

✓ Buffered spatial hold-out



Neighbors inside the buffer
are excluded from training



THE KEY DESIGN CHOICE

Every test station also has its close neighbors (within 5 km) removed from training — an inter-station audit found 24 station pairs closer than that. This is the single most defensible decision in the thesis.

- CHOOSING WHERE SENSORS GO

Greedy, on purpose

- At each step, place the sensor that closes the **biggest remaining gap**
- Chosen over swarm-based optimization (PSO) for two reasons: **transparency** and a **provable guarantee**
 - A planner can see exactly why each sensor was recommended — no black box to trust
 - For this class of coverage objective, greedy search is mathematically guaranteed to land within a **constant factor of the true optimum** — not just a heuristic hope

Greedy · transparent, provably near-optimal

PSO · possibly better, harder to explain

- THE DATA

Five public datasets, five years

SMHI air quality

SMHI weather

CORINE land cover

EU-DEM elevation

GEOSTAT population

- 2020–2024, 1,827 days of daily records
- All public and reproducible — no custom hardware, no proprietary data
- Common map projection (EPSG:3035) across every source

• DATA REALITY

34 stations. 11 usable.

34

total stations with records

19

pass completeness for one pollutant

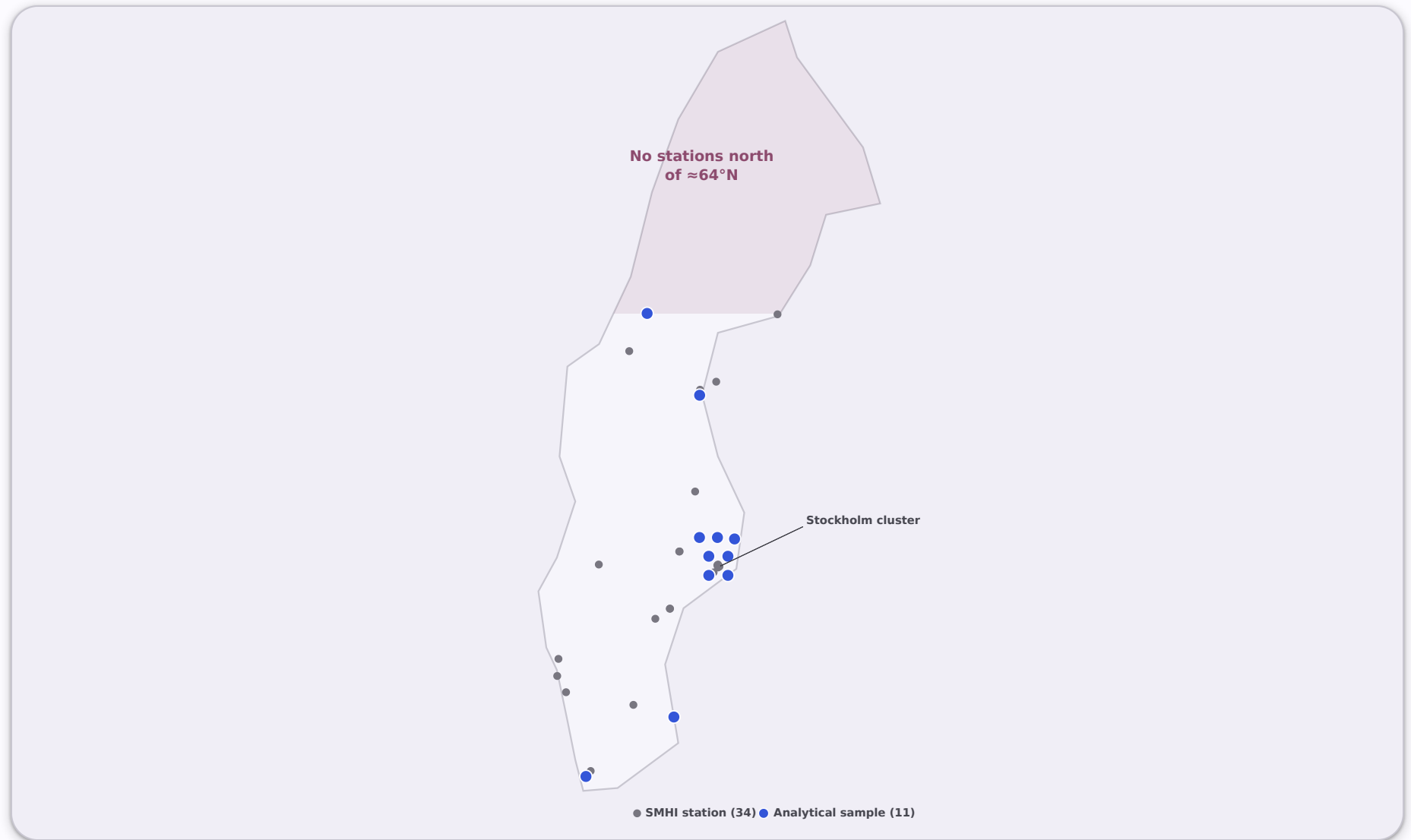
11

pass for both pollutants — the core sample

A 90% daily-completeness bar was applied per station, per pollutant. Only 11 stations clear it for *both* PM2.5 and NO2 — these 11 carry every validation fold, every decay curve, in the thesis.

● WHERE THE DATA COMES FROM

The network on the map



34 SMHI stations across Sweden, plotted by real coordinates — the north is almost entirely uncovered.

• ANSWERING RQ1

Accuracy, pollutant by pollutant

• **NO2 — Random Forest wins**

Mean RMSE ($\mu\text{g}/\text{m}^3$): **RF 7.27** vs IDW 8.90 vs LUR 15.33

• **PM2.5 — distance-averaging wins**

Mean RMSE ($\mu\text{g}/\text{m}^3$): **IDW 2.33** vs RF 3.56 vs LUR 5.87



BUFFERED SLOO — STEPS

- 1 Select a station, withhold it as the test point.
- 2 Draw the 5 km buffer around it.
- 3 Drop that station and any other station inside the buffer from training.
- 4 Train on what's left.
- 5 Predict at the withheld station, record its RMSE.

Every RMSE above was measured this way.

• BEHIND THE NUMBERS

Choosing the decay curve: AIC picks the winner

POLLUTANT	MODEL	BEST-FIT FORM	AIC	BIC
NO2	RF	log	60.68	61.47
NO2	LUR	log	83.51	84.30
NO2	IDW	log	63.38	64.18
PM2.5	RF	exponential	25.33	26.12
PM2.5	LUR	log	58.78	59.58
PM2.5	IDW	exponential	38.55	39.35



DECAY CURVE FITTING — STEPS

- 1 Test 3 curve shapes: log, power-law, exponential
- 2 Fit each with OLS → get **a** and **b**
- 3 Pick the best shape using AIC (lowest score wins)
- 4 Check the sign of **b** → confirms error grows with distance
- 5 Use IDW's curve (clearest trend) to find the threshold
- 6 Take 50% of real mean concentration = acceptable error

Not an assumed shape — a tested one.

n = 11 stations in every fit. Full candidate comparison — all three forms per combination, with fitted coefficients — is Table 7 in the thesis.

• ANSWERING RQ2

How far can an estimate be trusted?

≈64 km

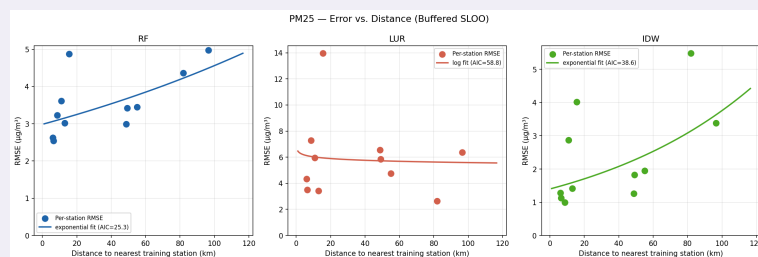
PM2.5 threshold (50% criterion)

≈6 km

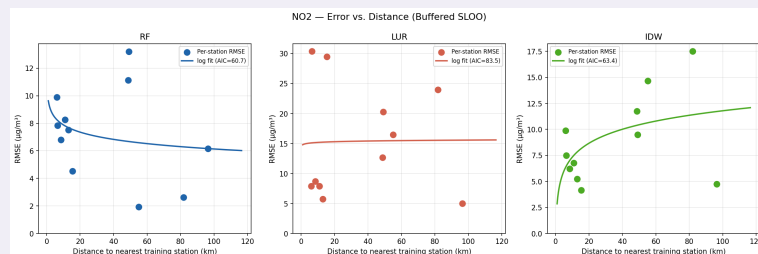
NO2 threshold (50% criterion)

Estimated prediction-distance thresholds under the adopted 50% RMSE criterion — not fixed reliability limits.

PM2.5



NO2



PM2.5 degrades gradually. NO2 degrades fast — tied tightly to nearby traffic.

• ANSWERING RQ3

20 sensors, national coverage

PM2.5 — the placement objective

~~10.9%~~ → **67.2%**

PM2.5 land coverage

~~38.8%~~ → **71.4%**

Urban area coverage

4 / 7

unmonitored regions served

NO2 — same 20 sensors, checked against its own stricter 6 km threshold

~~0.2%~~ → **0.6%**

NO2 land coverage

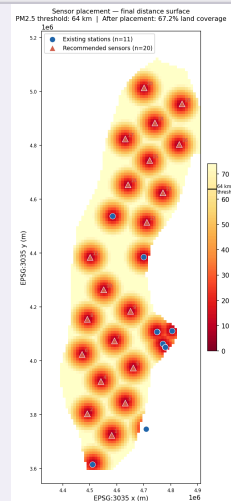
6.1%

NO2 urban coverage — unchanged

Each of the 20 placements picked, one at a time, whichever spot closed the biggest remaining gap in PM2.5 coverage — the same sensors were not chosen with NO2 in mind.

● WHERE THE SENSORS GO

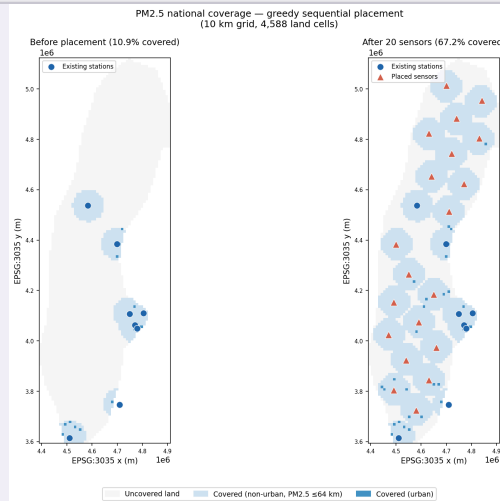
20 recommended locations



Spread from Götaland in the south, through Svealand, to Norrland in the north.

● THE COVERAGE GAIN

Before and after



Grey = beyond the 50%-criterion PM2.5 threshold. Blue = within reliable range of a station or sensor.

- SO WHAT

Two pollutants, two planning problems

- A PM2.5-sized deployment leaves NO2 exposure near traffic effectively unmonitored between stations
- Municipalities need to treat the two thresholds as **separate constraints**, not a shared budget
- Four newly served regions — Jönköping, Örebro, Dalarna, Norrbotten — are concrete near-term priorities

- WHERE TO PRESS

What this doesn't yet resolve

- Only **11 stations** carry every validation fold, decay curve, and threshold in the thesis
- The 5 km exclusion buffer is justified by station clustering, not the true spatial-autocorrelation range
- The evaluation grid (10 km) is coarser than the NO2 threshold itself (6 km)
- Straight-line distance stands in for road-network distance — more physically right for NO2, but not computable from this network
- The 50% criterion itself is a modelling choice, not a fixed rule — sweeping it 20–80% swings post-placement PM2.5 coverage from 2.9% to 94.7%

• THE CONTRIBUTION

One sentence



A reproducible method, built entirely on public data, that turns “how accurate is our estimate at distance X” into a ranked list of where to put the next sensor — demonstrated at the scale of an entire country.

Not claimed: an operational deployment plan

Not claimed: a regulatory compliance tool

- WHAT'S NEXT

Future work

- Validate against a denser reference network, or replicate in another country
- Jointly optimize placement for **both** pollutants instead of PM2.5 alone
- A finer national grid — fine enough to actually resolve the NO2 threshold
- Periodic recalibration as traffic patterns shift (e.g. vehicle electrification)

Thank you

Questions?

Spatial Estimation of Air Quality at Unmonitored Locations Using Machine Learning: Implications for IoT Sensor Network Design in Swedish Cities — Abdalazeez Asaad